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Zoom System for a Wash Light

Abstract

An optical system and luminaire are provided. The optical system includes a light source, a zoom lens module, and a positive power optical device that emits an output light beam. The light source emits one or more substantially parallel light beams. The zoom lens module includes a converging lens and a tubular light spill shield fixedly coupled to the zoom lens module. The converging lens receives the one or more light beams and emits a converging light beam, which passes through he tubular light spill shield to an aperture at its front end. The aperture is located adjacent to a focus of the converging lens. The positive power optical device receives the converging light beam. The light source and the positive power optical device are fixed in the optical system and the zoom lens module moves between them along an optical axis of the optical system.

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Background/Summary

TECHNICAL FIELD OF THE DISCLOSURE

[0001] The disclosure generally relates to automated luminaires, and more specifically to a zoom system for use in an automated wash light luminaire.

BACKGROUND

[0002] Luminaires with both manual and automated remotely controllable functionality are well known in the entertainment and architectural lighting markets. Such products are commonly used in theatres, television studios, concerts, theme parks, night clubs, and other venues. A typical automated luminaire provides control, from a remote location, of the output intensity, color, and other functions of the luminaire, and may allow an operator to control such functions for many luminaires simultaneously. Many automated luminaires additionally or alternatively provide control from the remote location of other parameters such as position, focus, zoom, beam size, beam shape, and/or beam pattern of light beam(s) emitted from the luminaire. A wash light is a type of luminaire which provides a soft edged light output typically used to illuminate performers and scenery.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] For a more complete understanding of the present disclosure and the advantages thereof, reference is now made to the following description taken in conjunction with the accompanying drawings in which like reference numerals indicate like features and wherein:

[0004] FIG. **1** presents a schematic view of a first optical system according to the disclosure.

[0005] FIGS. **2**A, **2**B, and **2**C present schematic views illustrating the zoom functionality of the optical system of FIG. **1**.

[0006] FIG. **3** presents a second optical system according to the disclosure.

[0007] FIG. **4** presents a third optical system according to the disclosure.

[0008] FIG. **5** presents a fourth optical system according to the disclosure.

[0009] FIG. **6** presents an exploded diagram of a fifth optical system according to the disclosure.

[0010] FIG. **7** presents a view of a luminaire according to the disclosure.

SUMMARY

[0011] In a first embodiment, an optical system includes a light source, a zoom lens module, and a positive power optical device configured to emit an output light beam from the optical system. The light source is configured to emit one or more substantially parallel light beams. The zoom lens module is optically coupled to the light source and includes a converging lens and a tubular light spill shield. The converging lens is configured to receive the one or more substantially parallel light beams emitted by the light source and to emit a converging light beam. The tubular light spill shield has a back end adjacent to a front surface of the converging lens and a front end comprising an aperture. The aperture is located adjacent to a focus of the converging lens. The tubular light spill shield is configured to pass the converging light beam. The converging lens and the tubular light spill shield are fixedly coupled to the zoom lens module. The positive power optical device is configured to receive the converging light beam and to emit the output light beam from the optical system. The light source and the positive power optical device are mounted in fixed positions in the optical system and the zoom lens module is configured to move along an optical axis of the optical system between the light source and the positive power optical device.

[0012] In some such embodiments the zoom lens module further comprises one or more fly-eye lens arrays, each comprising a plurality of converging lenslets. A first fly-eye lens array of the one or more fly-eye lens arrays is configured to receive the one or more substantially parallel light beams emitted by the light source and a second fly-eye lens array of the one or more fly-eye lens arrays is configured to emit an integrated converging light beam to be received by the converging lens. The one or more fly-eye lens arrays are fixedly coupled to the zoom lens module.

[0013] In a second embodiment, a luminaire includes a head and power circuits configured to provide electrical power to electrical circuits of the head. The head includes a light source that is configured to emit one or more substantially parallel light beams. The head further includes a zoom lens module optically coupled to the light source. The zoom lens module includes a converging lens, configured to receive the one or more substantially parallel light beams emitted by the light source and to emit a converging light beam, and a tubular light spill shield having a back end adjacent to a front surface of the converging lens and a front end comprising an aperture. The aperture is located adjacent to a focus of the converging lens. The tubular light spill shield is configured to pass the converging light beam. The converging lens and the light spill shield are fixedly coupled to the zoom lens module. The head still further includes a positive power optical device configured to receive the converging light beam and to emit an output light beam from the head. The light source and the positive power optical device are mounted in fixed positions in the head and the zoom lens module is configured to move along an optical axis of the head between the light source and the positive power optical device.

DETAILED DESCRIPTION

[0014] Preferred embodiments are illustrated in the figures, like numerals being used to refer to like and corresponding parts of the various drawings.

[0015] FIG. **1** presents a schematic view of a first optical system **100** according to the disclosure. The optical system **100** is a wash light optical system. Light is emitted from a light engine **105**. The light engine **105** comprises a plurality of LED emitters **104** mounted in an array on a substrate **102**. The LED emitters **104** may be of a single color (such as white) or may be in a plurality of colors. In various embodiments, the LED emitters **104** may be configured to be controllable as a single group, as multiple groups, or individually. In some embodiments, one or more of the LED emitters **104** has a primary optical system comprising a reflector, a total internal reflection (TIR) lens, or other suitable optical device. In some embodiments, the light engine **105** may contain laser light sources rather than (or in combination with) the LED emitters **104**. While the light engine **105** comprises a plurality of LED emitters **104**, in other embodiments, the light engine **105** comprises a single light source that emits a single collimated (or substantially parallel) beam.

[0016] Each LED emitter **104** is associated with a corresponding pair of collimating lenslets on lens arrays **106** and **108**. Each LED emitter **104** is optically coupled to and optically aligned with its corresponding collimating lenslet on the lens array **106**. Each collimating lenslet on the lens array **106** is optically coupled to and optically aligned with its corresponding collimating lenslet on the lens array **108**. That is, light from each of the LED emitters **104** passes first through its corresponding collimating lenslet on the lens array **106**, and then through its corresponding collimating lenslet on the lens array **108**.

[0017] Light rays of light beams from each LED emitter **104** and its collimating lenslets are substantially parallel (i.e., collimated). In some embodiments the term “substantially parallel” means that the half cone angle of light beams exiting the collimating lenslets on the lens array **108** is 10° (10 degrees) or less. In other embodiments, “substantially parallel” means that the half cone angle may be as low as 5° or as high as 20°. The LED emitters **104**, the substrate **102**, the collimating lens array **106** and the collimating lens array **108** may be assembled (or fixedly coupled) so as to form the light engine **105** as a unitary component.

[0018] Although the lens arrays **106** and **108** are shown in FIG. **1** as two separate substrates, in other embodiments, the lens arrays **106** and **108** may be fabricated on opposite sides of a single (common) substrate. In still other embodiments, the lens arrays **106** and **108** and their substrate(s) may be molded from a material comprising glass or a transparent polymer. In still further embodiments, the lens arrays **106** and **108** may be fabricated from multiple individual collimating lenslets.

[0019] The collimated and substantially parallel light beams emitted by the LED light engine **105** pass through a zoom lens module **115**. The zoom lens module **115** includes a converging lens **114**, an aperture **116**, and optional fly-eye lens arrays **110** and **112**. The fly-eye lens arrays **110** and **112** may be referred to as homogenizing or integration lens arrays. Each of the fly-eye lens arrays **110** and **112** comprises a plurality of converging lenslets. The fly-eye lens array **110** (a first fly-eye lens array) is configured to receive the collimated and substantially parallel light beams emitted by the LED light engine **105**. The fly-eye lens array **112** (a second fly-eye lens array) is configured to emit an integrated converging light beam to be received by a back surface of the converging lens **114**.

[0020] The individual fly-eye lenses of the fly-eye lens arrays **110** and **112** are smaller in diameter than the individual lenslets of the lens arrays **106** and **108**. Each collimated and substantially parallel light beam emitted by a lenslet illuminates a plurality of fly-eye lenses. Some fly-eye lenses may be illuminated by light beams from a plurality of adjacent lenslets. The lenslets of the lens arrays **106** and **108** may be separated from each other on their substrates. In contrast, the lenses of a fly-eye lens array are tightly packed together with little or no flat substrate between them. Such lenses may be hexagonal, in order to pack tightly in an hexagonal array.

[0021] The fly-eye lens arrays **110** and **112** are configured, along with converging lens **114**, such that the beam originating from each individual LED emitter **104** illuminates the aperture **116** (or gate) of the optical system **100**. Individual beams from the LED emitters **104** overlap in the aperture **116** before passing through further optical devices. The overlapping beams form an even, soft-edged beam. The aperture **116** may be a physical device (e.g., the aperture **116** as shown in FIG. **1**) or may be 'virtual' (i.e., a narrow region of the optical system where the beams from the LED emitters **104** overlap). The converging lens **114**, the aperture **116**, and the optional fly-eye lens arrays **110** and **112** are fixedly coupled to each other to form the zoom lens module **115** as a unitary component, which moves relative to the LED light engine **105** and an output lens **120**, as described in more detail below.

[0022] The optional fly-eye lens arrays **110** and **112** and the converging lens **114** are configured to overlap the individual light beams from the LED emitters **104** onto the aperture **116**. This overlap is configured to more fully integrate brightness variations between the light beams, homogenize colors of the light beams, and produce a combined light beam with a smoother illumination profile and a more fully homogenized color at the aperture **116**.

[0023] Although the fly-eye lens arrays **110** and **112** are shown in FIG. **1** as fabricated on two separate substrates, in other embodiments, the fly-eye lens arrays **110** and **112** may be fabricated on opposite sides of a single (common) substrate. In still other embodiments, the fly-eye lens arrays **110** and **112** and their substrate(s) may be molded from a material comprising glass or a transparent polymer. In still further embodiments, the fly-eye lens arrays **110** and **112** may be fabricated from multiple individual collimating lenslets.

[0024] The light beam that exits the aperture **116** (an output light beam) is a diverging beam that passes through (and is collimated by) the output lens **120**. The output lens **120** of the optical system **100** is a single Fresnel lens. In other embodiments, the output lens **120** may be a group of lenses, a pebble convex lens, a standard planoconvex lens, a searchlight-style reflector system, or other positive power optical device.

[0025] As described above, both the light engine **105** and the output lens **120** are mounted in fixed positions within the optical system **100**, however the zoom lens module **115** is configured to move along an optical axis **121** between the light engine **105** and the output lens **120**. Because the light beams emitting from light engine **105** are substantially parallel, the position of the zoom lens module **115** relative to, and its distance from, the light engine **105** does not substantially alter the character of the lights beam entering the zoom lens module **115**. As such, as the zoom lens module **115** moves along the optical axis **121**, it receives substantially the same light beams emitted by the light engine **105**.

[0026] FIGS. **2**A, **2**B, and **2**C present schematic views illustrating the zoom functionality of the optical system **100** of FIG. **1**. FIG. **2**A shows the optical system **100** in a first configuration, where the zoom lens module **115** is positioned close to the light engine **105**. In the first configuration, a light beam **122**A emitted by the output lens **120** has a narrow beam angle. In some embodiments, this narrow beam angle will be between 5° and 10°. FIG. **2**B shows the optical system **100** in a second configuration, where the zoom lens module **115** is positioned in an intermediate position between the light engine **105** and the output lens **120**. In the second configuration, a light beam **122**B emitted by the output lens **120** has an intermediate beam angle. Finally, FIG. **2**C shows the optical system **100** in a third configuration, where the zoom lens module **115** is close to the output lens **120**. In the third configuration, a light beam **122**C emitted by the output lens **120** has a wide beam angle. In some embodiments this wide beam angle will be between 50° and 60°. As discussed above, in all three configurations, because the light emitted from the light engine **105** is substantially parallel, it enters the zoom lens module **115** substantially unchanged as the zoom lens module **115** moves along the optical axis **121**.

[0027] FIG. **3** presents a second optical system **300** according to the disclosure. The optical system **300** includes the light engine **105** and the output lens **120** of the optical system **100**, as well as a zoom lens module **315**. The zoom lens module **315** includes a converging lens **314** and optional fly-eye lens arrays **310** and **312**, similar to those described for the optical system **100**.

[0028] The zoom lens module **315** further includes a tubular light spill shield **317**. The tubular light spill shield **317** is a hollow tubular structure that tapers from a back end adjacent to a front surface of the converging lens **314** down to a front end containing, or forming, the aperture **316**. The tubular light spill shield **317** has an axis that is coaxial with (or extends parallel to) the optical axis **121**. The light shield **317** is configured to pass a light beam emitted from the converging lens **314**.

[0029] In some embodiments, the tubular light spill shield **317** may have a black or matte internal surface, so as to inhibit reflection of stray light beams. In other embodiments, the light shield **317** may have a reflective, specular, or non-specular, internal surface so as to capture and redirect stray light beams. The tubular light spill shield **317** is fixedly coupled to the other components of, and moves with, the zoom lens module **315**. In some embodiments a diffuser is mounted in the aperture **316**, to further homogenize a light beam emitted by the converging lens **314**.

[0030] FIG. **4** presents a third optical system **400** according to the disclosure. The optical system **400** includes the light engine **105** and the output lens **120** of the optical system **100**, as well as a zoom lens module **415**. The zoom lens module **415** includes a converging lens **414** and optional fly-eye lens arrays **410** and **412**, similar to those described for the optical systems **100** and **300**. The zoom lens module **415** further includes a tubular light spill shield **417** and an aperture **416** similar to those described for the optical system **300**, as well as a light mixing chamber **418** and a diffuser **419**. The light mixing chamber **418** is coupled at a first end to the aperture **416**. The light mixing chamber **418** is a hollow tubular structure of any cross-sectional shape that may be straight or may taper. An internal surface of the light mixing chamber **418** may be reflective, either specular or non-specular, and operate to further homogenize an intensity profile and/or color of a light beam emitted by the aperture **416**. A second end of the light mixing chamber **418** includes the diffuser **419**, which further aids in homogenization of a mixed converging light beam emitted by the light mixing chamber **418**. A homogenized converging light beam emitted by the diffuser **419** illuminates the output lens **120**. The light mixing chamber **418** and the diffuser **419** are fixedly coupled to, and move with the other components of, the zoom lens module **415**.

[0031] FIG. **5** presents a fourth optical system **500** according to the disclosure. The optical system **500** includes the light engine **105** and the output lens **120** of the optical system **100**, as well as the zoom lens module **415** of the optical system **400**. The optical system **500** further includes a tubular light baffle **522**. The tubular light baffle **522** is a hollow tubular structure of any cross-sectional shape that may be straight or may taper and has an axis that extends parallel to the optical axis **121**. The internal surface of tubular light baffle **522** may be reflective, either specular or non-specular, so as to reflect and capture stray light emitted from the light engine **105** towards the zoom lens module **415**. The substantially parallel light beams emitted from the light engine **105** pass through hollow tubular light baffle **522** unobstructed. In some embodiments, the tubular light baffle **522** is fixedly coupled to the other components of, and moves with, the zoom lens module **415**. In other embodiments, the tubular light baffle **522** is fixedly coupled to the other components of the light engine **105** and does not move. In still other embodiments, the tubular light baffle **522** comprises a plurality of nesting, telescoping tubes that extend or contract in length as the zoom lens module **415** moves along the optical axis **121**. In such embodiments, a first end of the tubular light baffle **522** is fixedly coupled to the light engine **105** and a second end of the tubular light baffle **522** is fixedly coupled to the zoom lens module **415**.

[0032] FIG. **6** presents an exploded diagram of a fifth optical system **600** according to the disclosure. The optical system **600** includes a light engine **605** that comprises LED emitters **604** on a substrate **602**. The optical system **600** further includes lens arrays **606** and **608** and a zoom lens module **615**. The zoom lens module **615** comprises fly-eye lens arrays **610** and **612**, a converging lens **614**, a tubular light spill shield **617**, a light mixing chamber **618**, a diffuser **619**, and an output lens **620**. The diffuser **619** emits a diverging light beam that passes through (and is collimated by) the output lens **620**. The components of the optical system **600** function in similar ways to corresponding components of the optical systems **100**, **300**, **400**, and **500**.

[0033] FIG. **7** presents a view of a luminaire **750** according to the disclosure. The luminaire **750** comprises a head **752**, in which is mounted an optical system according to the disclosure (e.g., the optical system **600**). Also mounted in the head **752** are electrical circuits **754** (not visible in FIG. **7**). The electrical circuits **754** are configured to provide electrical power to the light engine **605** and to control circuits and motors configured to move the zoom lens module **615**. The luminaire **750** is a fixed position luminaire whose pan/tilt orientation is set by manually positioning the head **752** in a stirrup **756**.

[0034] In other embodiments, a luminaire comprises a head in which is mounted an optical system according to the disclosure. In such embodiments, the head is rotatably mounted in a motorized yoke assembly and configured for remotely controlled rotation about a tilt axis. The yoke assembly is rotatably mounted to a base and configured for remotely controlled rotation about a pan axis. In some such embodiments, the base may include power circuits configured to provide electrical power to electrical circuits of one or more of the base, the yoke assembly, and the head.

[0035] While only some embodiments of the disclosure have been described herein, those skilled in the art, having benefit of this disclosure, will appreciate that other embodiments may be devised which do not depart from the scope of the disclosure. While the disclosure has been described in detail, it should be understood that various changes, substitutions and alterations can be made hereto without departing from the spirit and scope of the disclosure.

Claims

1. An optical system comprising: a light source, configured to emit one or more substantially parallel light beams; a zoom lens module optically coupled to the light source, the zoom lens module comprising: a converging lens, configured to receive the one or more substantially parallel light beams emitted by the light source and to emit a converging light beam; and a tubular light spill shield having a back end adjacent to a front surface of the converging lens and a front end comprising an aperture, the aperture located adjacent to a focus of the converging lens, the tubular light spill shield configured to pass the converging light beam, wherein the converging lens and the tubular light spill shield are fixedly coupled to the zoom lens module; and a positive power optical device configured to receive the converging light beam and to emit an output light beam from the optical system, wherein the light source and the positive power optical device are mounted in fixed positions in the optical system and the zoom lens module is configured to move along an optical axis of the optical system between the light source and the positive power optical device.

2. The optical system of claim 1, wherein the zoom lens module further comprises: one or more fly-eye lens arrays, each comprising a plurality of converging lenslets, wherein: a first fly-eye lens array of the one or more fly-eye lens arrays is configured to receive the one or more substantially parallel light beams emitted by the light source and a second fly-eye lens array of the one or more fly-eye lens arrays is configured to emit an integrated converging light beam to be received by the converging lens; and the one or more fly-eye lens arrays are fixedly coupled to the zoom lens module.

3. The optical system of claim 1, wherein the aperture of the tubular light spill shield comprises a first diffuser.

4. The optical system of claim 1, wherein the tubular light spill shield comprises a black internal surface.

5. The optical system of claim 1, further comprising a light mixing chamber fixedly coupled to the zoom lens module and configured to receive the converging light beam emitted from the aperture of the tubular light spill shield and to emit a mixed converging light beam.

6. The optical system of claim 5, further comprising a second diffuser, fixedly coupled to the zoom lens module and configured to receive the mixed converging light beam and to emit a homogenized converging light beam to be received by the positive power optical device.

7. The optical system of claim 1, further comprising a tubular light baffle having a back end located adjacent to the light source and a front end located adjacent to a back surface of the converging lens, the tubular light baffle configured to pass the substantially parallel light beams emitted by the light source.

8. The optical system of claim 7, wherein the tubular light baffle comprises a reflective internal surface configured to reflect and capture stray light emitted from the light source.

9. The optical system of claim 7, wherein a first end of the tubular light baffle is fixedly coupled to the light source, a second end of the tubular light baffle is fixedly coupled to the zoom lens module, and the tubular light baffle comprises a plurality of nesting, telescoping tubes that extend or contract in length as the zoom lens module moves away or toward the light source.

10. A luminaire comprising: a head comprising: a light source, configured to emit one or more substantially parallel light beams; a zoom lens module optically coupled to the light source, the zoom lens module comprising: a converging lens, configured to receive the one or more substantially parallel light beams emitted by the light source and to emit a converging light beam; and a tubular light spill shield having a back end adjacent to a front surface of the converging lens and a front end comprising an aperture, the aperture located adjacent to a focus of the converging lens, the tubular light spill shield configured to pass the converging light beam, wherein the converging lens and the tubular light spill shield are fixedly coupled to the zoom lens module; and a positive power optical device configured to receive the converging light beam and to emit an output light beam from the head, wherein the light source and the positive power optical device are mounted in fixed positions in the head and the zoom lens module is configured to move along an optical axis of the head between the light source and the positive power optical device; and power circuits configured to provide electrical power to electrical circuits of the head.

11. The luminaire of claim 10, wherein the zoom lens module further comprises: one or more fly-eye lens arrays, each comprising a plurality of converging lenslets, wherein a first fly-eye lens array of the one or more fly-eye lens arrays is configured to receive the one or more substantially parallel light beams emitted by the light source and a second fly-eye lens array of the one or more fly-eye lens arrays is configured to emit an integrated converging light beam to be received by the converging lens; and the one or more fly-eye lens arrays are fixedly coupled to the zoom lens module.

12. The luminaire of claim 10, wherein the aperture of the tubular light spill shield comprises a first diffuser.

13. The luminaire of claim 10, wherein the tubular light spill shield comprises a black internal surface.
14. The luminaire of claim 10, further comprising a light mixing chamber fixedly coupled to the zoom lens module and configured to receive the converging light beam emitted from the aperture of the tubular light spill shield and to emit a mixed converging light beam.
15. The luminaire of claim 14, further comprising a second diffuser, fixedly coupled to the zoom lens module and configured to receive the mixed converging light beam and to emit a homogenized converging light beam to be received by the positive power optical device.
16. The luminaire of claim 10, further comprising a tubular light baffle having a back end located adjacent to the light source and a front end located adjacent to a back surface of the converging lens, the tubular light spill shield configured to pass the substantially parallel light beams emitted by the light source.
17. The luminaire of claim 16, wherein the tubular light baffle comprises a reflective internal surface configured to reflect and capture stray light emitted from the light source.
18. The luminaire of claim 16, wherein a first end of the tubular light baffle is fixedly coupled to the light source, a second end of the tubular light baffle is fixedly coupled to the zoom lens module, and the tubular light baffle comprises a plurality of nesting, telescoping tubes that extend or contract in length as the zoom lens module moves away or toward the light source.
19. The luminaire of claim 10, further comprising: a yoke assembly, wherein the head is rotatably mounted in the yoke assembly for rotation about a tilt axis.
20. The luminaire of claim 19, further comprising: a base, wherein the yoke assembly is rotatably mounted to the base for rotation about a pan axis.